

Adam Martinez

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Education

University of California, Santa Cruz - Computer Science, M.S (Expected 2028)

University of California, Santa Cruz - Computer Science, B.S (2026)

- Relevant Courses: Computer Systems and C Programming, Machine Learning, Artificial Intelligence, Functional Programming, Computer Architecture, Software Development

Skills:

- Python, C, C++, ROS2, OpenCV, PyTorch, Tensorflow, Git, Docker, CARLA, Linux, Gazebo
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Experience

Autonomous Vehicles Assistant AI Researcher @ AIEA Research Lab, UCSC ([Link](#) | [Github](#))

- Worked on a team of industry professionals to assess the viability of graph search for the AV problem, with an emphasis on passenger safety and efficient obstacle avoidance.
- Compared graph search algorithms like A*, D*, and D* Lite for autonomous vehicle path planning in urban environments, demonstrating progressively lower replanning costs and decreased cumulative path computation with D* Lite compared to A*.
- Tuned models as part of experimentation in reinforcement learning for the autonomous vehicle problem.
- Used LIDAR sensor information for obstacle detection and avoidance.
- Built models and test scenarios in the CARLA simulator environment.

Reinforcement Learning Research Intern @ Baskin Engineering, UCSC ([Link](#))

- Designed neural networks using PyTorch and TF workflows.
- Assessed the performance of various RL approaches for controlling urban autonomous vehicles, including AC and DQN based approaches.
- Established the efficiency of policy-optimized actor-critic models over DQN for more consistent reward gains during early learning, and resilience to over-fitting.
- Presented findings to industry professionals in a research conference held at the University of California, Santa Cruz.
- Worked with state translation from sensor data and BEV state representations to optimize positional and routing computations.

Autonomous Pan-Tilt Camera with Facial Recognition

- Designed, planned, and built a real-world pan-tilt camera robot, automatically detecting and following human faces.
- Used closed-loop control tracking logic implemented for autonomous, continuous motion.
- Utilized an OpenCV-PyTorch-ROS2 workflow to train a CNN capable of detecting face position within images, and autonomously actuating motors toward face center.
- Integrated modular ROS software onto a Raspberry Pi controller hardware and microservo actuators.

Instruction Coordinator @ Association of Computing Machinery ([Link](#))

- Lectured on web development, networking and cybersecurity, and AI/ML development.